

Multiple Choice

1. **Answer:** A

Options: A) 3 gallons; B) 5 gallons; C) 8 gallons; D) 2 gallons; E) 1 gallon

Note: Correct option: A - 3 gallons

2. **Answer:** A

Options: A) 14.15 M; B) 16.20 M; C) 15.00 M; D) 12.75 M; E) 9.95 M

Note: Correct option: A - 14.15 M

3. **Answer:** B

Options: A) 3.0×10^{-3} ; B) 2.0×10^{-3} ; C) 1.7×10^{-1} ; D) 1.5×10^{-2} ; E) 2.0×10^{-1}

Note: Correct option: B - 2.0×10^{-3}

3. **Answer:** A

Options: A) $1.2\text{dm}^3 \text{ mol}^{-1}$; B) $1.35\text{dm}^3 \text{ mol}^{-1}$; C) $2.0\text{dm}^3 \text{ mol}^{-1}$; D) $0.8\text{dm}^3 \text{ mol}^{-1}$; E) $1.8\text{dm}^3 \text{ mol}^{-1}$

Note: Correct option: A - $1.2\text{dm}^3 \text{ mol}^{-1}$

4. **Answer:** E

Options: A) 0.12 molal; B) 0.05 molal; C) 0.4 molal; D) 0.2 molal; E) 0.1 molal

Note: Correct option: E - 0.1 molal

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '67.2 liters'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 7, neutral. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 0.15 volts. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.